LOKÁLNE A GLOBÁLNE EXTRÉMY

1) Lokálne extrémy = defininované na celom D(f)

Príklad: Nájdite lokálne extrémy f(x,y) ...

2) Globálne extrémy = funkcia ohraničená množinou

Príklad: Nájdite extrémy funkcie f(x,y) ohraničenú množinou M

3) Nelineárna optimalizácia = funkcia má väzby
= Lagrangeova funkcia

Príklad: Nájdite extrémy funkcie f(x,y) ak podmienka väzby (ak sú premenné viazané)

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2) GLOBÁLNE EXTRÉMY: $f(x,y)$ $M = \{[x,y] \in R^2 \mid x=0,\ y>x\}$

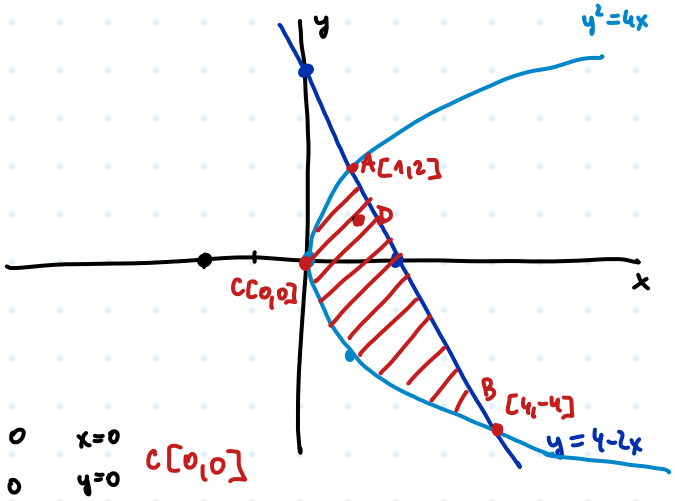
1 KROK: nakreslíme graf ohraničenia ⇒ nájdeme KRITICKÉ BODY (priesečníky)
→ Lokálne extrémy

2 KROK: funkcie ohraničenia a dosadíme do f(x,y)
→ f' → KANDIDÁTOV NA EXTRÉMY

3. KROK: Body dosadíme do f(x,y) najväčšia hodnota MAX
najmenšia hodnota MIN

Príklad: Nájdite extrémy funkcie $f(x,y) = 16 - x^2 - y^2$ na množine $M = \{[x,y] \in R^2 \mid y^2 = 4x,\ y = 4-2x\}$.

1 KROK:



$y^2 = 4x = y = 4-2x$

$(4-2x)^2 = 4x$

Lokálne extrémy:

$f'_x = -2x = 0 \quad x=0$

$f'_y = -2y = 0 \quad y=0$

C[0,0]

2. KROK: $y^2 = 4x$ → dosadím do f(x,y) → $f(x,y) = 16 - x^2 - 4x$

↳ $f' = -2x - 4$

$-2x - 4 = 0$

$-2x = 4$

$x = -2$

$y^2 = 4 \cdot (-2)$

$y^2 = -8$ ✗

$y = 4 - 2x$ → $f(x;y)$

$f(x;y) = 16 - x^2 - (4-2x)^2$

$= 16 - x^2 - (16 - 16x + 4x^2)$

$= 16 - x^2 - 16 + 16x - 4x^2$

$= -5x^2 + 16x$

$f' = -10x + 16$

$-10x + 16 = 0$

$10x = 16$

$x = \frac{16}{10} = \frac{8}{5}$

$y = 4 - 2\cdot\frac{8}{5} = 4 - \frac{16}{5} = \frac{4}{5}$

$D\left[\frac{8}{5}, \frac{4}{5}\right]$

3 KROK:

$A[1,2] = 16 - 1^2 - 2^2 = 11$

$B[4,-4] = 16 - 4^2 - (-4)^2 = -16$ MIN

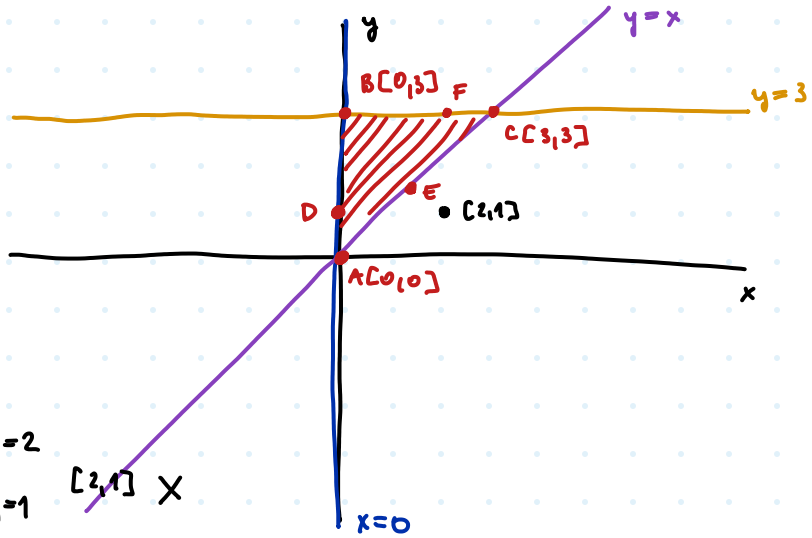
$C[0,0] = 16 - 0^2 - 0^2 = 16$ MAX

$D\left[\frac{8}{5}, \frac{4}{5}\right] = 16 - \left(\frac{8}{5}\right)^2 - \left(\frac{4}{5}\right)^2 = \frac{64}{5} \sim 12,\ldots$

Funkcia $f(x;y)$ má v bode $[4,-4]$ minimum s hodnotou -16 a v bode $[0,0]$ maximum s hodnotou 16.

Príklad Nájdite extrémy funkcie $f(x;y) = x^2 - 4x + y^2 - 2y$ na množine $M = \{[x;y] \in \mathbb{R}^2 \mid x \geq 0,\ y \geq x,\ y \leq 3\}$

1. KROK:



Lokálne extrémy:

$f'_x = 2x - 4$

$f'_y = 2y - 2$ $\Rightarrow$ $x = 2$, $y = 1$

$[2,1] \times$

$x = 0$

2. KROK:

$x = 0$ → $f(x;y)$

$f(x;y) = 0^2 - 4\cdot 0 + y^2 - 2y$

$= y^2 - 2y$

$f' = 2y - 2$

$2y = 2$

$y = 1$

$D[0,1]$

$y = x$

$f(x;y) = x^2 - 4\cdot x + x^2 - 2x$

$= 2x^2 - 6x$

$f' = 4x - 6$

$4x = 6$

$x = \frac{6}{4} = \frac{3}{2}$

$E\left[\frac{3}{2}, \frac{3}{2}\right]$

$y = 3$

$f(x;y) = x^2 - 4x + 9 - 6$

$= x^2 - 4x + 3$

$f' = 2x - 4$

$2x - 4 = 0$

$2x = 4$

$x = 2$

$F[2,3]$

3 KROK:

$A[0,0] = 0$

$B[0,3] = 3$ MAX

$C[\frac{3}{2},3] = 0$

$D[0,1] = -1$

$E[\frac{3}{2},\frac{3}{2}] = -\frac{9}{2}$ MIN

$F[2,3] = -1$

Funkcia $f(x,y)$ má v bode $[0,3]$ maximum s hodnotou 3 a v bode $[\frac{3}{2},\frac{3}{2}]$ minimum s hodnotou $-\frac{9}{2}$.

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3) Lagrangeova metóda

↳ máme funkciu a k nej máme väzbu

$f(x,y)$ $g(x,y)$

1. KROK: Lagrangeova rovnica

$$L(x,y) = f(x,y) + \underbrace{\lambda}_{\text{lambda}} \cdot g(x,y) + \lambda_2 \cdot g_2(x,y) + \lambda_3 g_3(x,y) \dots$$

2 KROK:

$$\left.\begin{aligned} L'x &= 0 \\ L'y &= 0 \\ g(x,y) &= 0 \end{aligned}\right\} \Rightarrow \text{BODY}$$

3. KROK:

$$\text{Hessián} = \begin{vmatrix} L''xx & L''xy \\ L''yx & L''yy \end{vmatrix}$$

4. KROK: λ dosadíme do Hessiánu

> 0 MÁME EXTRÉM → $L''xx$ >0 MIN, <0 MAX

< 0 NEMÁME EXTRÉM

$= 0$ nevieme

Príklad: Nájdite extrémy funkcie $f(x,y) = 4-4x-2y$ ak premenné sú viazané podmienkou $g(x,y) = x^2+y^2-25$

1. KROK: $L(x,y) = 4-4x-2y + \lambda(x^2+y^2-25)$

2 KROK:

$L'x = -4 + 2x\lambda = 0$ (1)

$L'y = -2 + 2y\lambda = 0$ (2)

$x^2+y^2-25 = 0$ (3)

(1) $2x\lambda = 4$

$x\lambda = 2$

$x = \frac{2}{\lambda}$

(2) $2y\lambda = 2$

$y\lambda = 1$

$y = \frac{1}{\lambda}$

(3) $\left(\frac{2}{\lambda}\right)^2 + \left(\frac{1}{\lambda}\right)^2 = 25$

$\frac{4}{\lambda^2} + \frac{1}{\lambda^2} = 25$

$\frac{5}{\lambda^2} = 25$

$\lambda^2 = \frac{1}{5}$

$\lambda = \pm\sqrt{\frac{1}{5}}$

$\lambda = \sqrt{\frac{1}{5}}$ $\quad x = \frac{2}{\lambda} = \frac{2}{\sqrt{\frac{1}{5}}}$ $\quad y = \frac{1}{\sqrt{\frac{1}{5}}}$

$A\left[\frac{2}{\sqrt{\frac{1}{5}}}, \frac{1}{\sqrt{\frac{1}{5}}}\right]$ MIN

$\lambda = -\sqrt{\frac{1}{5}}$

$B\left[\frac{-2}{\sqrt{\frac{1}{5}}}, \frac{-1}{\sqrt{\frac{1}{5}}}\right]$ MAX

3. KROK:

$$\begin{vmatrix} L''_{xx} & L''_{xy} \\ L''_{yx} & L''_{yy} \end{vmatrix} = \begin{vmatrix} 2\lambda & 0 \\ 0 & 2\lambda \end{vmatrix} = 4\lambda^2$$

$L'_x = -4 + 2x\lambda$

$L'_y = -2 + 2y\lambda$

4 KROK λ dosadíme do Hessiány

$\lambda = \sqrt{\frac{1}{5}} \Rightarrow 4\frac{1}{5} = \frac{4}{5} > 0$ MÁME EXTRÉM

$\lambda = -\sqrt{\frac{1}{5}} \Rightarrow 4\frac{1}{5} = \frac{4}{5} > 0$ MÁME EXTRÉM

$L''_{xx} = 2\lambda$ $\quad \lambda = \sqrt{\frac{1}{5}} \Rightarrow 2\cdot\sqrt{\frac{1}{5}} > 0$ MIN

$\lambda = -\sqrt{\frac{1}{5}} \Rightarrow -2\sqrt{\frac{1}{5}} < 0$ MAX

Funkcia $f(x;y)$ má v bode $\left[\frac{2}{\sqrt{\frac{1}{5}}}, \frac{1}{\sqrt{\frac{1}{5}}}\right]$ minimum s hodnotou _____ a v bode $\left[\frac{-2}{\sqrt{\frac{1}{5}}}, \frac{-1}{\sqrt{\frac{1}{5}}}\right]$ maximum s hodnotou _____ .

Príklad Nájdite extrémy funkcie $f(x;y) = 2 - x - 2y$, väzba $g(x;y) = 2x^2 + y^2 - \frac{9}{2}$

1. KROK: $L(x;y) = 2 - x - 2y + \lambda\left(2x^2 + y^2 - \frac{9}{2}\right)$

2 KROK:

$L'_x = -1 + \lambda(4x) = 0$ (1)

$L'_y = -2 + \lambda(2y) = 0$ (2)

$2x^2 + y^2 - \frac{9}{2} = 0$ (3)

(1) $4\lambda x = 1$ $\quad x = \frac{1}{4\lambda}$

(2) $2y\lambda = 2$ $\quad y = \frac{1}{\lambda}$

13) $2\left(\frac{1}{4x}\right)^2 + \left(\frac{1}{\lambda}\right)^2 = \frac{9}{2}$

$\frac{2}{16x^2} + \frac{1}{x^2} = \frac{9}{2}$

$\frac{18}{16x^2} = \frac{9}{2}$

$16x^2 \cdot 9 = 18 \cdot 2$

$x^2 = \frac{1}{4}$

$\lambda = \sqrt{\frac{1}{4}} = \frac{1}{2}$ x, y $A\left[\frac{1}{2}, 2\right]$ MIN

$\lambda = -\frac{1}{2}$ x, y $B\left[-\frac{1}{2}, -2\right]$ MAX

3. krok:

$$\begin{vmatrix} 4x & 0 \\ 0 & 2\lambda \end{vmatrix} = 8\lambda^2$$

4 krok:

$\lambda = \frac{1}{2}$ ⤳ $8\left(\frac{1}{2}\right)^2 = 2 > 0$ MÁME EXTRÉM

$\lambda = -\frac{1}{2}$ ⤳ $8\left(-\frac{1}{2}\right)^2 = 2 > 0$ MÁME EXTRÉM

$l'_{xx} = 4\lambda$

$\lambda = \frac{1}{2} \Rightarrow 2$ MIN

$\lambda = -\frac{1}{2} \Rightarrow -2$ MAX

Funkcia $f(x,y)$ má v bode $\left[\frac{1}{2}, 2\right]$ minimum s hodnotou -2.5 a v bode $\left[-\frac{1}{2}, -2\right]$ maximum s hodnotou 6.5